

Comments on the Bose Condensation of Phonons in the Biological Systems

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Abstract

Following the conjecture of Fröhlich that the Bose condensation of phonons is the dominant biological effect, Wu and Austin derived from the Fröhlich Hamiltonian an expression for the relaxation time which is inversely proportional to the temperature by using the technique of Green's function. In this paper, we have generalized the Wu and Austin expression by improving their approximations with the superconducting canonical transformation and obtained the relaxation time depending exponentially on the inverse temperature. Moreover, we have checked our expression by using a different method based on the "continued fraction expansion" and also employed this technique to derive the dielectric constant for this system.

Introduction

Fröhlich [1-6] proposed that the phonon condensation may be possible in the biological systems because of their extraordinary dielectric properties. The dipoles of the biosystem generate longitudinal coupled oscillations to produce a narrow phonon energy band. Fröhlich [1-6] showed from the rate equation that such a phonon system will undergo Bose Condensation if it is coupled to a thermal bath system and an energy pump with a pumping rate exceeding a critical value. Wu and Austin [7] showed that this phenomenon of Bose condensation in the biosystem can be demonstrated from the fundamental quantum mechanical arguments. Furthermore, Wu and Austin [8] derived an expression for the relaxation time in terms of the phonon-bath coupling coefficient and estimated the numerical value for it.

In this paper we have derived an expression for the dielectric constant in terms of the same phonon-bath coupling coefficient. The calculations have been based on the "continued fraction expansion" technique developed by one of the authors (Paul [9]), and provided an estimate of this coefficient from the available experiment data.

Theory

Wu and Austin [7] assumed for the biosystem the following Hamiltonian of the

superconductors introduced by Fröhlich [10]:

$$H = \sum_i \hbar \omega_i a_i^\dagger a_i + \sum_k \hbar \Omega_k b_k^\dagger b_k + \sum_p \hbar \Omega_p P_p^\dagger P_p + \sum_{ijk} \hbar (\chi a_i^\dagger a_j b_k^\dagger + \chi^* a_j a_i^\dagger b_k) + \sum_{pi} (\xi P_p a_i^\dagger + \xi^* P_p^\dagger a_i), \quad (1)$$

where each of the normal modes of the oscillation of the biosystem is represented by the creation and the destruction operators $a_a^\dagger$ and a_i , respectively. Similar operators ($b_k^\dagger, b_k$) and ($P_p^\dagger, P_p$) are assigned for the rest of the biosystem which are the bath and the pump, respectively. The χ is the coupling coefficient which is responsible for the interaction between the phonons and the bath.

Rate of change of the number of quanta in the i th mode N_i is given by the Heisenberg equation

$$\frac{dN_i}{dt} = \frac{1}{i\hbar} [N_i, H]; \quad (2)$$

and the expectation value of Eq. (2) is

$$\left\langle \frac{dN_i}{dt} \right\rangle = \left\langle \psi(t) \left| \frac{dN_i}{dt} \right| \psi(t) \right\rangle. \quad (3)$$

Using the perturbation method and the commutation relationship of the boson operators, Wu and Austin [7] derived the rate equation given by

$$\left\langle \frac{dN_i}{dt} \right\rangle = S_i - \Phi(T, \omega_i) [\langle N_i \rangle e^{\beta \omega_i} - (1 + \langle N_i \rangle)] - \sum_j \Lambda(T, \omega_i, \omega_j) [\langle N_i \rangle (1 + \langle N_j \rangle) e^{\beta \omega_i} - \langle N_j \rangle (1 + \langle N_j \rangle) e^{\beta \omega_j}], \quad (4)$$

where we have $\beta = \hbar/kT$, S is the rate of supply of energy from external source to the i th mode, and Φ depends on the coupling coefficients other than χ .

For the stationary state, we have

$$\left\langle \frac{dN_i}{dt} \right\rangle = 0. \quad (5)$$

Using Eqs. (4) and (5) one may solve for the average occupation number of each phonon state and obtain an expression of the form

$$\langle N_i \rangle \approx (e^{\beta(\omega_i - \mu_i)} - 1)^{-1}, \quad (6)$$

where μ_i is defined as

$$e^{-\beta \mu_i} = (\Phi(T, \omega_i) + \sum_j \Lambda(T, \omega_i, \omega_j) (1 + \langle N_j \rangle)) \times (\Phi(T, \omega_i) + \sum_j \Lambda(T, \omega_i, \omega_j) \langle N_j \rangle e^{\beta \omega_j})^{-1} \quad (7)$$

and

$$\Lambda(T, \omega_i, \omega_j) = (2\pi/\hbar)|\chi|^2 e^{-\beta(\omega_i - \omega_j)} \quad (8)$$

So if we have $\chi = 0$, $\mu_i = 0$, no condensation can take place.

Bose condensation is possible, however, if and only if

$$\omega_i > \mu_i \geq 0. \quad (9)$$

From the rate equation it is obvious that the Bose condensation in the biosystem depends upon the coupling coefficient χ . Wu and Austin [11] have shown that this coupling coefficient dominates the other ones that appear in Eq. (1). Therefore, they employed Green's function technique [8] to derive an expression for the relaxation time τ_1 in terms of χ only

$$\tau_1 = \hbar^3 \pi \nu'^3 / |\chi|^2 k T \gamma \Delta, \quad (10)$$

where ν' is the speed of sound, k is the Boltzman constant, T is the absolute temperature, Δ is the width of the energy band, and γ is a positive quantity less than 1. From Eq. (10) we estimated a numerical value for $|\chi|$ as 4×10^{-2} by using $\tau_1 \approx 10^{-10}$ s, $\nu' \approx 300$ m/s and $\gamma \Delta \approx 10^6$ as given in Ref. 8.

Measurement of χ from the Dielectric Constant

Since oscillators involved are dipolar in nature it is easy to see that they should leave a remarkable effect upon the dielectric properties of the system. If a medium composed of dipolar entities is subjected to a rapidly changing electromagnetic field the dipoles will oscillate with the field but if they are coupled to a bath, then the response will be modified by this coupling. This modification will appear in the form of a phase lag which is related to the imaginary part of the dielectric constant $\epsilon''(\omega_0)$ and therefore $\epsilon''(\omega_0)$ is related to the parameter χ . The precise details of this calculation are based upon the continued fraction technique developed by Paul [9] involving a two-body Green's function. These details will be published elsewhere but for the present we are providing a simplified version of the final equation in order to estimate χ from experimental measurement of the dielectric constant.

The expression for the dielectric constant derived by us is given by

$$\epsilon''(\omega_0) = \frac{4\pi^3 \hbar^2 \rho |\mu|^2 \omega_0 (N_i - N_j)^2}{|\chi|^2 k T \Gamma(\omega_0)}, \quad (12)$$

where ρ is the density of the material, $|\mu|$ is the mean dipole moment, ω_0 is the resonant frequency, and $\Gamma(\omega_0)$ is the linewidth. Since the relaxation time is inversely proportional to $\Gamma(\omega_0)$ and in our derivations the latter contains exponentials of $(-\hbar\omega_0/kT)$, thus the relaxation time will be proportional to the exponent of temperature's inverse.

Discussion and Results

Supekar and Bandopathyaya [12] provided some experimental data for cholesterol

at and around two resonant frequencies which are given below in the SI units

$$\begin{aligned}\rho &= 7.5 \text{ kg/m}^2, \quad T = 300^\circ \text{K}, \\ \omega_{01} &= 0.63 \text{ MHz}, \quad \Gamma(\omega_{01}) = 0.92 \text{ MHz}, \\ \epsilon''(\omega_{01}) &= 0.270 \text{ F/m}, \\ \omega_{02} &= 2.19 \text{ MHz}, \quad \Gamma(\omega_{02}) = 3.16 \text{ MHz}, \\ \epsilon''(\omega_{02}) &= 0.265 \text{ F/m}.\end{aligned}$$

Using the above values, we obtained from our equation (12) $|\chi/\hbar| = 5 \times 10^{-3}$ (experimental accuracy is 2%) for both resonant frequencies. This shows that the coupling coefficient indeed does not depend on the frequency. This value of $|\chi/\hbar|$ agrees satisfactorily with the value found from formula (10) of Wu and Austin which is $|\chi/\hbar| \approx 4 \times 10^{-2}$.

Detailed measurement of the dielectric constant of the biosystem in microwave frequencies are in progress in our EPR laboratory.

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